About the job

Role: AIOps Engineer

About Parspec

Parspec is building the AI and digital infrastructure for the construction materials supply chain.

Construction is a $15 trillion industry, yet the systems that underpin the buying and selling of materials remain fragmented, manual, and disconnected. Distributors and rep agencies rely on spreadsheets, PDFs, phone calls, and siloed tools to find new products and quote and manage projects; creating delays, errors, and margin erosion across the supply chain.

Parspec is an AI-native platform that powers how construction products are discovered, bought, and sold. Trusted by more than 300 MEP distributors and rep agencies, Parspec helps project-driven businesses bid faster, win more work, and operate more profitably. By combining product intelligence, AI-powered workflows, and a connected ecosystem, Parspec is laying the foundation for a more intelligent, efficient construction supply chain.

Founded in 2021 and headquartered in San Mateo, California, Parspec has raised $31 million from leading deep-tech and construction-technology investors.

The Opportunity

We are looking for an experienced **AIOps / LLMOps Engineer** to help design, deploy, and manage the AI infrastructure that powers Parspec's next-generation AI systems.

This role will focus on building and maintaining **scalable, secure, and observable AI platforms** that support generative AI applications and document intelligence workflows. You will work on self-hosted large language models, asynchronous inference pipelines, and production-grade ML infrastructure on AWS.

You will collaborate closely with AI researchers, product teams, and backend engineers to ensure reliable and efficient deployment of AI systems across Parspec's platform.

Preferred location: **Bengaluru**, with regular in-office collaboration.

What You Will Achieve And Key Responsibilities

AI Infrastructure & LLM Platform Development

- Design and build document AI platforms powered by generative AI, leveraging asynchronous architectures for scalable inference.

- Implement event-driven and queue-based systems to support elastic scaling and non-blocking AI workflows.
- Architect and maintain self-hosted LLM infrastructure using tools such as vLLM or Ollama on Kubernetes or EC2 with GPU orchestration.

LLM Operations & Model Governance

- Manage production systems for LLM serving, inference pipelines, and AI workflow orchestration.
- Implement LLM gateways and routing systems (e.g., LiteLLM, Portkey) to ensure proper model usage and governance.
- Develop guardrails and monitoring systems to reduce hallucinations, misuse, and unsafe outputs in generative AI systems.

Observability & AI System Monitoring

- Implement end-to-end observability for AI/ML pipelines using distributed tracing and monitoring tools.
- Monitor AI system health using platforms such as OpenTelemetry, AWS X-Ray, Prometheus, and Grafana.
- Track performance metrics including latency, token usage, inference quality, and model drift.

ML Platform & Workflow Management

- Manage machine learning workflows using tools such as MLflow, Kubeflow, or SageMaker MLFlow setups.
- Enable experiment tracking, model versioning, and deployment pipelines for production AI systems.
- Work closely with engineering teams to integrate AI workflows into scalable backend systems.

Security & Infrastructure Optimization

- Implement AI platform security controls including Bedrock Guardrails, KMS encryption, IAM least-privilege policies, VPC endpoints, and CloudTrail auditing.
- Optimize AWS infrastructure—including Bedrock, SageMaker, and EKS—for cost efficiency, performance, and reliability.
- Ensure production AI systems maintain high availability and security standards.

Why This Matters

Generative AI systems are only as powerful as the infrastructure that supports them. Building reliable, scalable AI platforms, especially those that serve complex document intelligence workloads, requires deep expertise in distributed systems, observability, and model operations.

At Parspec, this infrastructure powers AI systems that extract and organize construction product data at scale. Your work will enable the next generation of document AI and generative workflows that make critical industry information accessible, searchable, and actionable.

Who You Are

Required Qualifications

- Strong experience with AWS cloud infrastructure including services such as EC2, Lambda, S3, EKS, Bedrock, Step Functions, API Gateway, EventBridge, and SQS/SNS.
- Experience building ML infrastructure using Infrastructure-as-Code tools such as Terraform or CloudFormation.
- Hands-on experience deploying and operating LLM serving infrastructure using platforms such as vLLM or Text Generation Inference.
- Experience managing vector databases and retrieval systems such as Pinecone, PGVector, or Weaviate.
- Strong experience designing event-driven or asynchronous systems using queues (SQS, Kafka) and micro-batching patterns.
- Experience implementing observability and monitoring for distributed AI systems using tools such as ELK, Prometheus, Grafana, and OpenTelemetry.
- Strong programming experience in Python, including frameworks such as FastAPI and asynchronous programming patterns (asyncio).
- Experience with Docker, Kubernetes, and CI/CD pipelines using tools such as GitHub Actions or ArgoCD.

Experience

- 5+ years of experience in MLOps, LLMOps, AIOps, or DevOps supporting machine learning or AI systems.
- Proven track record building production generative AI systems with high availability and scalability.
- Experience deploying self-hosted LLMs on AWS infrastructure and building production-grade document AI platforms.
- Experience operating AI systems with >99.9% uptime and cost-efficient infrastructure management. JobRequirements_AIOPS_Engineer

Preferred Qualifications

- Bachelor's or Master's degree in Computer Science, Data Science, or a related technical field.
- Experience working with multimodal LLMs or agent-based AI workflows.
- Familiarity with cost-optimized inference hardware and infrastructure such as AWS Trainium.

What we offer

- Competitive salary and benefits, including family insurance coverage, free health teleconsultations, and learning/upskilling budgets
- Equity in the company
- Flexible hours and a hybrid work setup
- Unlimited PTO
- Opportunity to grow with a fast-scaling company transforming a large market
- Preferred Location: San Mateo, with regular in-office presence

Join Us

At Parspec, we recognize that traditional job descriptions don't always capture the full range of your unique abilities—and that's perfectly okay. You may not meet every requirement, but if you bring a mix of experiences, fresh perspectives, and a passion that aligns with our mission, we want to hear from you!

The Parspec team believes that varied backgrounds drive better outcomes and fuel innovation. We are a team of self-starters that lead from every seat. We think big, set a standard of excellence and are committed to diversity and a discrimination-free workplace. We welcome applicants from all walks of life to join us and help shape the future at Parspec.